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Tree Risk Assessment Report

Site

Rock Lane Farm, Liscombe Park, Soulbury, Bucks, LU7 0JL

Client

Thameside Construction Company Limited

Prepared by

Patrick Stileman BSc(Hons), MICFor, Dip. Arb (RFS), M.Arbor.A

Date

14th November 2011

Ref

TI12101101

1 INTRODUCTION

- 1.1 I am Patrick Stileman, Director of Patrick Stileman Ltd. I am acting on instruction of the client, Thameside Construction Company Limited. I have qualifications and experience in arboricultural consultancy, and I have given details of this in Appendix 3.
- 1.2 **Brief:** We are instructed by the client to provide an assessment of the risk posed by trees located within the area highlighted on a plan provided to us, at Rock Lane Farm, Liscombe Park, Soulbury, Bucks, LU7 0JL. We are to provide recommendations for work to the trees (including suggested timeframes) required in order to bring the risk that they pose to a level which we consider to be defensible. I have included a copy of the plan provided showing the area within which trees are to be inspected, as Figure 1.
- 1.3 **Report scope:** This report relates only to trees located within the highlighted area of the plan provided. The possible effect that these trees may have to structures through changes in soil volume is not included in this report. The report is of a preliminary nature only; a detailed inspection was not undertaken on any tree, nor were any samples of plant or soil taken and sent to laboratory for analysis.
- 1.4 **Tree identification:** The approximate location of the trees has been shown on the site plan included as Figure 2 which is based on the Ordnance Survey plan that we purchased. The position of trees shown on the site plan is indicative only. I attached a metal numbered tag to each individual tree inspected which corresponds to the number allocated in the survey schedule.
- 1.5 **Legal status of trees:** I was told by residents living at the site that trees are protected by a Tree Preservation Order (TPO). I have not seen a copy of the TPO and do not know at this stage which of the trees is protected.
- 1.6 **Site visit:** I carried out a site visit on 2nd November 2011.

2 METHOD OF INSPECTION

- 2.1 Inspection of the trees was based on a technique developed by Claus Mattheck called Visual Tree Assessment (VTA). Practitioners of VTA use growth features on trees (body language) to interpret internal defects and provide an assessment of the likelihood of failure. My inspection was carried out from ground level. I did not carry out soil excavation or take root or soil samples for analysis. The conclusions that I have reached are based on an interpretation of my observations using my knowledge and experience.

- 2.2 I carried out an assessment of the risk that the trees pose by using a system called *Tree Hazard: Risk Evaluation and Treatment System* (THREATS). THREATS considers the three basic components of risk, being: the likelihood of failure; the nature of the land use within striking distance (the target); and the size of the tree, or part of tree in which failure is being considered. By scoring each of these and multiplying them together, a 'threat category' is determined which dictates whether or not remedial work is required, and if so within what time-scale. For a full description of the THREATS methodology, refer to Appendix 1.

3 SUMMARY OF RECOMMENDATIONS

3.1 Table 1: Summary of work required for risk management within three months

Tree No	Species	Work recommended
803	Lime	Remove large dead wood
809	Larch	Remove
813	Ash	Reduce crown by 25%, to be achieved by 2-3 metre reduction in height and spread
815	Larch	Remove large dead wood
816	Hawthorn	Remove broken, hung-up branch. Reduce crown on SW side by 30%
820	Lime	Remove large dead wood

3.2 Table 2: Summary of work required for risk management within one year

Tree No	Species	Work recommended
804	Walnut	Remove long lateral limb on north side
806	Larch	Reduce large dead wood
808	Lime	Remove large dead wood
810	Larch	Remove large dead wood
811	Larch	Reduce length of leaning section by approximately 4 metres. Remove large dead wood
812	Horse chestnut	Remove one low dead limb on north side

- 3.3 I recommend that, unless otherwise stated, all trees are re-inspected within three years.

4 STATUTORY PROTECTION

- 4.1 It is my understanding that trees at this site are protected by a TPO, however I have not seen the order and do not know which trees this applies to. For several trees I have recommended the removal of dead wood and this work is exempt from requiring consent under Section 198(6)(a) of the Town and Country Planning Act 1990 as amended (however, if relying on this exemption it is good practice to give the LPA at least 5 day's notice prior to carrying out the work). For all other work that I have recommended, written consent will be required from Aylesbury Vale District Council if the trees are protected. I can provide further assistance with this if required.

5 WILDLIFE

- 5.1 Nesting birds, bats and bat roosts are protected by law. It is the duty of the contractors to satisfy themselves prior to commencement that neither these species, nor any protected species shall be adversely affected by the proposed work. Contractors should have ecological policies detailing how work shall proceed in relation to protected species. I recommend that tree work is carried out in accordance with the Arboricultural Association Guidance Note 1 '*Bats in the Context of Tree Work Operations*'.

6 IMPLEMENTATION

- 6.1 **Implementation:** All work is to be carried out in accordance with BS3998 (2010) *Recommendations for tree work*. The contractors should be trained in the work that they are performing; carry public liability insurance (it is for the client to satisfy themselves that a suitable level of cover is held by the contractor; however £5 million is a minimum level generally considered to be acceptable); and undertake written risk assessments for the work being undertaken. We recommend that a certificate of insurance and site specific risk assessments should be seen by the client prior to the contractor commencing work. Some local authorities hold a list of approved contractors, though ideally the selected contractor should be approved by the Arboricultural Association. A list of Arboricultural Association approved contractors, can be viewed on line at www.trees.org.uk/contractors.htm, or call 01242 522152.

7 REPORT LIMITATIONS

- 7.1 Trees by their very nature will always pose a certain level of risk. This report is not intended to provide recommendations for the complete removal of risks from trees. The report is based on my assessment of the trees and provides recommendations for reasonable levels of management required in order to bring them to a level of risk which I consider to be defensible.

Patrick Stileman

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Chartered Arboriculturist. Arboricultural Association Registered Consultant

Figure 1

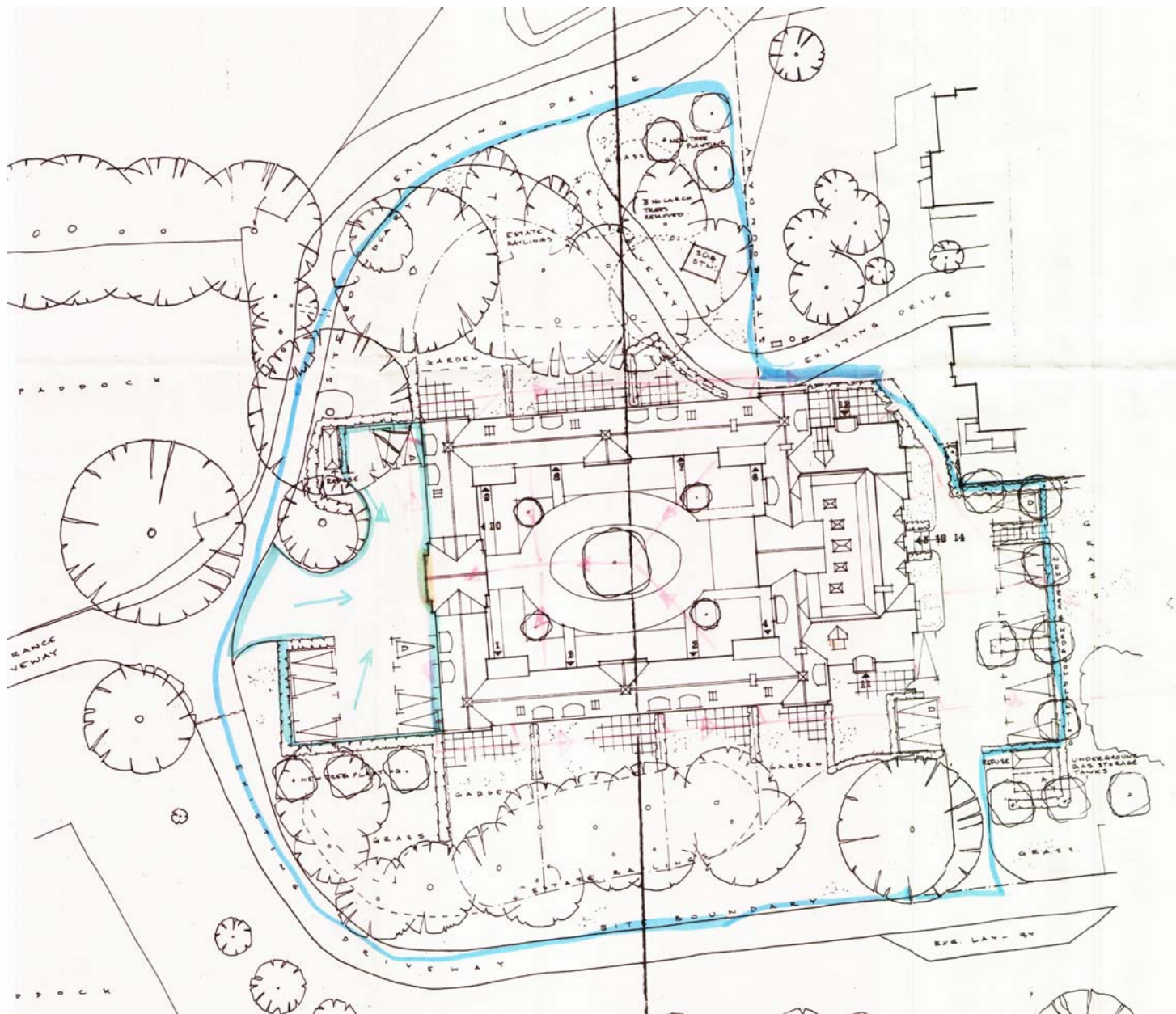




FIGURE 2

TREE LOCATION PLAN

SITE ADDRESS

Rock Lane Farm, Liscombe Park,
Soulbury, Bucks, LU7 0JL

CLIENT

Thameside Construction Company Ltd

JOB REF

TI12101101

DRAWING NO

TI12101101.01

DATE

14/11/2011

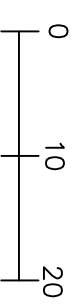


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KEY

● Approximate position of tree stem

SCALE



Metres

NOTE:

This drawing is based on the Ordnance Survey base plan. All tree positions have been estimated and are indicative only.

APPENDIX 1 THREATS METHODOLOGY

THREATS is an acronym for Tree Hazard: Risk Evaluation and Treatment System. It combines mathematical quantification with the tree inspectors assessment and interpretation of the significance of tree defects, to provide a consistent and systematic approach to the decision making process.

THREATS considers in turn the three components that inevitably combine to derive risk, being: the likelihood of failure; the target value (the likelihood of people, or objects of value being within striking range of the tree at any given time); and the likely impact of the defect being assessed if it were to fail.

Each of the three components described above are assessed, and awarded a numerical 'score'. These scores are derived in the following way:

- **Failure Score:** Using training and experience, the assessor identifies defects within the tree, and uses his or her judgement to assess the relevance of these defects. The assessor then chooses the defect with the greatest significance, and allocates one of five scores to that defect based on the likelihood and imminence of it causing failure.

Score	Probability of failure	Example defects
50	Imminent/Immediate	Uprooting; Extreme root loss; Collapsing structure, unimpeded hanging breaks
8	Probable/Soon	Altered exposure; Primary decay fungus; Severe inclusive bark/root loss, Fragile dead wood
2	Likely, foreseeable	Lapsed pollard; Overweight/subsiding limbs; Poor stem taper; Dieback
.8	Potentially with time	Early development of inclusive bark; Robust dead wood
0	Unlikely ever	Tree generally free of defects, or insignificant defects only

- **Target score:** This describes the relative value of the land use within likely striking distance of the component part affected by the selected defect. It also takes into account the likelihood of human occupancy at any given time. The assessor selects one of the following six scores:

Score	Value	Static target examples	Target occupancy examples
40	Very high	Building 24 hour use, railway	Constant vehicular traffic/busy playground
25	High	Building 12 hour use, ≥ 11 Kv power lines	Frequent vehicular traffic/constant pedestrian use
20	Medium	Building/structure occasional use, < 11 Kv lines	Peak times traffic/intermittent use, eg commuter run
15	Low	Garage, Summer house, Listed wall	Occasional traffic/sporadic use, eg slow country road
7	Very low	Unlisted wall, paving, garden features	Infrequently used access/public right of way/bridleway
0	None	Grass	Hardly ever used, eg remote path

- **Impact score:** The assessor makes a judgement on the likely consequences should the defect being assessed cause failure of the tree or part of tree in question if the target beneath it is occupied. The assessor selects one of the following four scores:

Score	Degree of harm and consequences (examples)	Agent: trees, mm, or branches, kg (size/weight for guidance only)		
10	Severe structural damage, vehicles crushed – passenger fatalities very probable	VL	> 750mm	> 500kg
6	Moderate structural/ severe vehicle damage – fatal/disabling injuries likely	L	350-750mm	50-500kg
4	Minor damage/probable disabling/hospitalising injury to pedestrians	M	100-350mm	10-50kg
1	Fragile objects destroyed, superficial/recoverable injury to pedestrians	S	< 100mm	< 10kg

The three scores derived from the methodology described above are multiplied together to provide a ‘hazard rating’. The hazard rating score will fall into one of seven ‘threat category’ score ranges, from 1 ‘insignificant’ to 7 ‘extreme’. The threat category that is finally reached determines whether or not remedial work is required, and a timescale in which any remedial work should be carried out, or the tree re-inspected.

The following is an extract from the THREATS survey sheet, indicating the score ranges (derived from a multiplication of the three scores listed above), the ‘threat category’ that these score ranges fall into, and whether or not intervention is required based on the threat category, with timescales.

Score range	Threat Category	Recommended action & Completion deadline
4000+	7- Extreme	Evacuate/prevent access to impact site, emergency call-out of contractors
2001-3999	6- Serious	Close site if practical; arrange for work to be completed within 7 days
1000-2000	5- Significant	Arrange for work to be completed within one month maximum
350-999	4- Moderate	Remediate within 3 months, re-inspect after gales in the meantime (Force 7+)
160-349	3- Slight	Re-inspect annually/after storms (Force 10+), expect to schedule work within 2 yrs
50-159	2- Minimal	Re-inspect within 3 yrs if adjacent to public access, schedule work as required
0-49	1- Insignificant	Re-inspect within 3 yrs if Target Score = 20, 25 or 40. Otherwise reassess within 5 years.

APPENDIX 2 TREE SCHEDULE

Explanatory notes

- **T no:** Tree number allocated to each tree during the survey. Each tree is prefixed with the letter T. Where trees form distinct groups in which I considered it unnecessary to select trees individually, the prefix G is given.
- **Species:** I base the species identification on visual observations and list the common English name, with the botanical name given in brackets the first time each new species is recorded.
- **Age class:** I estimate the approximate stage of the trees life, where Y = young, MA = middle-aged, M = mature, OM = over-mature or veteran.
- **Size class and ht:** The size of the tree is based on the trunk diameter. S = small (<100 mm) M = medium (100 – 350 mm) L = large (350 – 750 mm) and VL = very large (> 750 mm). The estimated height of the tree (in metres) is recorded
- **Condition, observations and defects:** A brief description of the trees condition, with the principal defects considered. The principal defect which is being scored for the purposes of THREATS is **highlighted in bold**, where applicable.
- **FS:** Failure score (See Appendix 1)
- **TS:** Target score (See Appendix 1)
- **IS:** Impact score (See Appendix 1)
- **Hazard Rating:** Score derived by multiplying FS, TS and IS.
- **Threat Cat:** Threat category from a range of 1 ‘insignificant’ to 7 ‘extreme’. See Appendix 1.
- **Recommendations:** Based on the threat category, the decision is made whether or not remedial work is required, and in what timescale if it is. Recommendations made are based on my knowledge and experience (see Appendix 3).
- **Urgency code:** Maximum timescale required for either the work recommended as stated in the previous column, or future re-inspections to be carried out.

APPENDIX 2 (continued)

Tree schedule

T no	Species	Age class	Size Class & Ht	Condition, Observations and defects	FS	TS	IS	Hazard Rating	Threat Cat.	Recommendations	Urgency code
801	Walnut (<i>Juglans regia</i>)	M	L 13	FAIR. Re-grown from past crown reduction. Spreading form with satisfactory crown structure. Old impact wounds (x2) on stem at ground level and at 1.5 metres on north side have good wound-wood development. Twin-stemmed from 1.2 metres. Union currently appears stable but strength could deteriorate in the future	0.8	20	6	96	2	Re-inspect	3 years
802	Walnut	M	L 10	FAIR. Tree has re-grown from past crown reduction. Old wound on SW side of stem is largely occluded but has caused internal hollowing. Wound and canker on lateral limb on SW side are potentially vulnerable but significant weight has been removed from the branch end and currently appears stable.	0.8	20	6	96	2	Re-inspect	3 years
803	Common lime (<i>Tilia x europaea</i>)	M	L 19	GOOD. Large, prominent tree. Moderate sized dead wood appears vulnerable, otherwise no defects seen of apparent structural significance	8	20	4	640	4	Remove large dead wood	3 months
804	Walnut	M	L 14	FAIR. Slender form, distorted from competition with Tree 803. Small dead wood of low significance. Long lateral limb on north side with high end-weight distribution and damage along its length appears vulnerable	2	20	4	160	3	Remove long lateral limb on north side	1 year
805	Hawthorn (<i>Crataegus monogyna</i>)	Tree was in an extremely hazardous condition at the time of the inspection, being in the process of failing. Arboricultural contractors were called out to the site immediately, and removed the tree on the day of the inspection									
806	European larch (<i>Larix decidua</i>)	M	L 16	GOOD. Moderate sized dead wood appears vulnerable, otherwise no defects seen of apparent structural significance	8	20	1	160	3	Remove large dead wood	1 year
807	Horse chestnut (<i>Aesculus hippocastanum</i>)	MA	L 16	GOOD. Small dead wood. Early indications of bacterial bleeding canker on limb on south side, currently of no structural significance	0.8	20	4	72	2	Re-inspect	3 years

APPENDIX 2 (continued)

Tree schedule

T no	Species	Age class	Size Class & Ht	Condition, Observations and defects	FS	TS	IS	Hazard Rating	Threat Cat.	Recommendations	Urgency code
808	Common lime	MA	L 16	GOOD. Moderate sized dead wood appears vulnerable, otherwise no defects seen of apparent structural significance	8	20	1	160	3	Remove large dead wood	1 year
809	European larch	M	M 15	POOR. Slender stem with low vitality. Basal wounds in two locations, one with apparent decay developing on the east side. Tapping the stem with a sounding mallet implies significant basal decay on the west side. Foreseeable hazard of whole tree failure	8	20	6	960	4	Remove	3 months
810	European larch	M	M 15	FAIR. Slight lean to the north. Moderate dead wood throughout, otherwise no defects seen of apparent structural significance	8	20	1	160	3	Remove large dead wood	1 year
811	European larch	M	M 15	FAIR. Moderate dead wood throughout. Upper section of tree has pronounced lean to the north making entire tree potentially vulnerable	2	20	6	240	3	Reduce length of leaning section by approximately 4 metres. Remove large dead wood	1 year
812	Horse chestnut	M	L 15	FAIR. Wound at base on south side is of low apparent significance. Short dead limb on north side , otherwise no defects seen of apparent structural significance	2	20	4	160	3	Remove one low dead limb on north side	1 year
813	Ash (<i>Fraxinus excelsior</i>)	M	VL 19	FAIR. Large tree with wide, spreading form. Decay seen in scaffold limb on the north side, and several degraded <i>Inonotus hispidus</i> fungal fruiting bodies have fallen from the tree and are lying on the ground surrounding it. Stub from past limb failure on the north side. Foreseeable hazard of major limb failure	8	20	6	960	4	Reduce crown by 25%, to be achieved by 2-3 metre reduction in height and spread	3 months
814	European larch	M	M 15	FAIR. Slender, suppressed tree with narrow stem. Sheltered by Tree 15. No defects seen of apparent structural significance	0	2	-	0	1	Re-inspect	3 years
815	Common lime	M	L 20	GOOD. Moderate sized dead wood through crown, otherwise no defects seen of apparent structural significance	8	20	4	640	4	Remove large dead wood	3 months

APPENDIX 2 (continued)
Tree schedule

T no	Species	Age class	Size Class & Ht	Condition, Observations and defects	FS	TS	IS	Hazard Rating	Threat Cat.	Recommendations	Urgency code
816	Hawthorn	M	L 11	FAIR-POOR. Low vitality. Heavily reduced on north side with pronounced imbalance to SW over adjacent garden. Twin-stemmed from near ground level. Large dead branch hanging in crown on north side	8	20	4	640	4	Remove broken, hung-up branch. Reduce crown on SW side by 30%	3 months
817	Horse chestnut	M	L 9	FAIR. Heavily reduced tree. Twin-stemmed from 1.5 metres with tight union on north side only. Dense ivy on stem. No defects seen of apparent structural significance	0	20	-	0	1	Re-inspect	3 years
818	European larch	M	M 19	FAIR. Slender tree with slight lean to the north-east. Minor wounds at base are of low apparent significance. Small dead wood throughout. Hammock attached to stem	0.8	20	6	96	2	Re-inspect	3 years
819	European larch	M	M 19	FAIR. Slender tree with slight lean to the north-east. Minor wounds at base are of low apparent significance. Small dead wood throughout. Hammock attached to stem	0.8	20	6	96	2	Re-inspect	3 years
820	Common lime	M	L 20	GOOD. Minor basal wounds of low significance. Moderate sized dead wood through crown , otherwise no defects seen of apparent structural significance	8	20	4	640	4	Remove large dead wood	3 months
821	Crab apple (<i>Malus sp</i>)	Y	S 4	GOOD. No defects seen of apparent structural significance	0	20	-	0	1	Re-inspect	3 years

APPENDIX 3.

Qualifications and experience of Patrick Stileman BSc(Hons), MICFor, Dip.Arb(RFS), M.Arbor.A

I am Patrick Stileman, director of Patrick Stileman Ltd Arboricultural Consultancy.

My qualifications in arboriculture are as follows:

National Certificate in Arboriculture *Nch(arb)*

The Arboricultural Associations Technicians Certificate *Tech.Cert (Arbor.A)*

The Royal Forestry Society's Professional Diploma in Arboriculture *Dip.Arb(RFS)*

In addition to the qualifications listed above which are specific to the field of arboriculture, I also hold an honours degree in Environmental Science *BSc(Hons)*.

I hold chartered status, being a Chartered Arboriculturist and professional member of the Institute of Chartered Foresters *MICFor*.

I am a registered consultant with the Arboricultural Association.

I am a trained expert witness, and hold the Cardiff University Bond Solon Expert Witness Certificate.

I am a member of the Royal Forestry Society.

I have been working within the arboricultural industry since 1994 and have been carrying out consultancy work since 2001. I am frequently instructed by professionals to provide advice and assistance relating to trees within the planning process, and I have a wide client base in this field including developers, architects, planning consultants, and Local Planning Authorities. I am experienced with providing arboricultural input in planning appeals as written representation, informal hearing and public local inquiry.

I am regularly instructed to assist with tree risk assessments, and to provide guidance relating to tree safety. Past clients for this work include Local Authorities, (notably St Albans District Council, Dacorum Borough Council, Wycombe District Council, Woking District Council, Hertfordshire and Surrey County Councils), schools, housing associations and private individuals.

I frequently provide advice in relation to alleged tree-related damage to buildings. Clients for this work are typically domestic homeowners, but have also included Hertfordshire County Council and Dacorum Borough Council. Other work that I undertake involves the provision of tree planting schemes; and advice relating to the general management of trees.

Prior to running my current consulting practice, I was a partner in an arboricultural contracting business in which I was involved with the practical aspect of organising, and execution of contract tree work.